22CST103 Problem Solving using C , Autumn 2025-2026, Department of CSE
Mid Term Exam MNIT Jaipur
Attempt all questions

Duration: 90 Minutes | Maximum Marks: 30

1. What will the output of the following code segments? [4*2]

A.
```
#include <stdio.h>
int main()
{
 int a=25;
 while(a<=27)
 {
   a++;
  if(a==27) continue;
  printf("%d", a);
 }
 return 0;
}
```

B.
```
#include<stdio.h>
int main()
{
  int MNIT = 1;
  switch (MNIT << (2 + MNIT))
  {
  default:
     printf("ECE");
  case 4:
     printf("C_programming");
  case 5:
     printf(" CSP");
  case 8:
     printf("CSE");
  }
 return 0;
 }
```

```
#include<stdio.h>
int main()
{
  int a=15, b=1,e,f;
 if(a<b?a:b)
 {
 e=a&b;
 printf("%d", e);
 }
 else
 {
 f=a^b;
 printf("%d",f);
 } return 0; }
```

D.
```
#include<stdio.h>
int main()
{
  int a, b = 5;
        a = -b--;
        printf("a = %d, b = %d", a, b);

 return 0;
}
```

2. Write a program in 'C' that takes input an integer number of seconds and print the equivalent time in hours, minutes and seconds. Recommended output format is something like 7322 seconds is equivalent to 2 hours 2 minutes 2 seconds. [3]

3. A palindrome number is a number that remains the same when its digits are reversed. Like **16461**. Write a program in 'C' to find whether a number is palindrome or not. [3]

4. Write a program in 'C' to print following pattern: [3]

 1
 1 2
 1 2 3
 1 2 3 4
 1 2 3 4 5

5. Write a program in 'C' to calculate the sum of 10 numbers. If user inputs negative number then negative number skips from calculation. (Note: use of goto or continue statement) [3]

or

A prime number is a natural number greater than 1 that cannot be formed by multiplying two smaller natural numbers. Example prime numbers in the range 1 to 30 are 2, 3, 5, 7, 11, 13, 17, 19, 23, and 29. Write a program in 'C' to display list of prime numbers from X to Y, where X and Y will be entered by user. [3]

Malaviya National Institute of Technology Jaipur
Department of Computer Science and Engineering
Mid Term Examination (Sept. 2024)
24AIT102 – Problem solving with C

Duration: 1.5 hrs

Max Marks: 30

Q1. State which of the following are true or false. Justify. **5 marks**

a) C operators are evaluated from right to left.

b) The following are all valid variable names:

i. _start#here

ii. Πmathsymbol

c) In a C programming switch statement, the default case is mandatory and must always be included.

d) Cache memory is faster but has less capacity compared to main memory (RAM).

e) Conditions written inside flowchart decision symbols are limited to only arithmetic operators (i.e., +, -, *, /, and %).

Q2. a) For a given array, write a program to find the equilibrium point. The equilibrium point in an array is an index (or position) such that the sum of all elements before that index is the same as the sum of elements after it. **1.5 marks**

b) Difference between **3 marks**

i. semantic error and syntactic error 1

ii. system software and application software

iii. compiler and interpreter

Q3. Convert the following as directed. **3+3+0.5 marks**

i. Decimal to binary, octal and hexadecimal

a. $(1792)_{10}$

b. $(9241.98)_{10}$

ii. Hexadecimal to binary and octal.

a. $(88BAE)_{16}$

b. $(4FB2)_{16}$

c. $(DC4.A7)_{16}$

iii. Find the binary equivalent of $(-12)_{10}$ in 2's complement.

Q5. a) For the given value of n, print the following patterns: **3 marks**

Eg. For n = 2 the pattern will be 2 2 1 1 2 1	Eg. For n = 3 the pattern will be 3 3 3 2 2 2 1 1 1 3 3 2 2 1 1 3 2 1

b) Identify and correct the errors in the following code. 3 marks

```
#include<stdio.h>
int main() {
Int x == 10;
Float y = 5.5
if (x > y);
    printf("x is greater than y")
else if (x = y)
    printf("x is equal to y");
return 0
}
```

Q6. Write a program to find out if a given year is a leap year or not, print the output to the terminal. A year is a leap year if "any one of" the following conditions are satisfied: 3 marks

1. The year is multiple of 400.
2. The year is a multiple of 4 and not a multiple of 100.

Q7. Predict the output of the following codes. 1x5 = 5 marks

a)

```
#include<stdio.h>
int main() {
int i = 0;
switch (i) {
case '0': printf("Mid
Term");
case '1': printf("End
Term");
break;
default: printf("Mid Term
and End Term");
}
return 0;}
```

b)

```
#include<stdio.h>
int main(){
int a = -10, b = 20;
if(a > 0 && b < 0)
   a++;
else if(a < 0 && b < 0)
   a--;
else if(a < 0 && b > 0)
   b--;
else
   b++;
printf("%d", a + b);
return 0;}
```

c)

```
#include <stdio.h>
int main() {
    int i = 2;
    if (i == 0)
        printf("Zero");
    else if (i = 1)
        printf("One");
    else if (i == 2)
        printf("MNITJ");
    else if (i == 3)
        printf("Three");
    else
        printf("Other");
    printf(",i = %d", i);
    return 0;}
```

d)

```
#include <stdio.h>
int main(){
int a = 6,b=6,c = 6;
printf("%d", (a == b)
+ (b == c) + (a ==
c));
return 0;}
```

e)

```
#include <stdio.h>
int main(){
  int i = 6, j = 10;
  j = i++ + ++i + i--;
  printf("%d,%d", i,
j);
  return 0;}
```

Malaviya National Institute of Technology Jaipur
Department of Computer Science & Engineering
CST-103 Problem Solving using C
MTE (22/09/23)

Max Marks : 30 **Time : 1.5 hours**

NOTE: Attempt all question. Use of calculator is not allowed.
Justify your answer in not more than 5 Lines.(Assume there is no typo mistake)

1) What is associativity and why is it required? (2)

2) Which conditional operator can form a logic equivalent to "if-else" statement? Explain with example. (2)

3) Let the initial values are x=5, a=7, b=8; Find the final value of 'x' for the following expressions (Justify your answer and show each step): (3)

i) **x += 2*3/2+ ++a-b-- +(5?3>2:2);**
ii) **x*= 3*(a++%x)/--a+(b/=5);**
iii) **x -= (a++ > b--) || (a++ > b--) || (a++ > b--);**

4) Ram has told Shyam that he is N days old. Write an algorithm and then convert it into C program so that Shyam can understand the age of Ram in terms of years, days and hours. (2+3 =5)

5) What will be the output for this code fragment? (2.5)

```
#include<stdio.h>
int main()
{
   int i=4, j= -1, k=0, w, x, y, z;
   w = i || j || k;
   x = i && j && k;
   y = i || j &&k;
   z = i && j || k;
   printf("%d, %d, %d, %d\n", w, x, y, z);
   return 0;
}
```

6) What will be the output for this code fragment? (2.5)

```
int main()
{ int a=2,b=3,c=4;
  if(b==2)
    a=10;
else
   c=10;
   printf("\na=%d \t b =%d \t c=%d",a,b,c);
return 0;

}
```

7) Which of the following operations are INCORRECT and why? (2.5)

a. int i = 35; i = i%5;
b. short int j = 255; j = j;
c. long int k = 365L; k = k;
d. float a = 3.14; a = a%3;

8) What will be the output for this code fragment? (2.5)

```
int main()

{ int a, b;
  const a1;
  a1=a+b;
  printf("%d",a1);
  return 0;

}
```

9) Complete the following program in such a way that when the user input the value 2 then it prints "Welcome" and If it enters "5" then it should display "I don't understand your input" (3)
#include<stdio.h>
void main() {
int check ;
switch (check)
{
 	
}
}
	10) Write the short notes on:- (5)
	(a). Primary and secondary memory
	(b). What is USB flash drive?
	(c). What is pre-processor?
	(d). Different type of programming errors
	(e) The intermediate files generated during the execution of a C program

"The expert in anything was once a beginner-Helen Hayes"